

SWYC Processing

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2026-02-24

Overview of SWYC Normative Scoring (Sheldrick et al., 2019)

Sheldrick et al., 2019: <https://pubmed.ncbi.nlm.nih.gov/31727860/>

The standard clinical application of the Survey of Well-being of Young Children (SWYC) relies on rigid, age-specific scoring thresholds to categorically flag children at risk for developmental delay. While useful in a rapid primary care setting, this approach artificially truncates the variance of developmental trajectories, limiting its utility for high-resolution statistical modeling.

To address this, this pipeline implements the continuous normative scoring methodology developed by Sheldrick et al. (2019) Using Item Response Theory (IRT), specifically a Graded Response Model, this approach translates raw SWYC item responses into continuous, age-standardized metrics.

Key outputs of this methodology include:

- **Developmental Age Estimation:** Rather than simply summing a score, this method uses established logistic curves to calculate the probability of a child's specific combination of responses across a continuous theoretical age range (0–72 months). The age at which the joint probability of their answers peaks is calculated as their **Developmental Age**.
- **Developmental Quotient (DQ):** By dividing the computed Developmental Age by the child's chronological age, we derive the DQ. This provides a standardized, continuous metric (ideally centered around 1.0) that quantifies a child's developmental progression relative to normative expectations.
- **Robustness to Missingness:** Because the IRT model evaluates the joint probability of the *provided* answers, rather than strictly penalizing for missing ones, it remains highly robust when individual questions are skipped.

This document outlines the extraction, cleaning, and scoring of the SWYC developmental milestone questionnaires from the electronic health record (EHR) found in Arcus. It calculates both standard clinical thresholds and research-grade normative developmental quotients (DQ).

For convenience, the relevant methodology from the supplement is shown below. Code was not provided by the authors to replicate this pipeline and thus this notebook is based only on the in-text description.

To calculate the DQ, developmental age is first estimated on the basis of parents' responses to each question. Specifically, an item response curve is generated for each question answered by using the equations above. Each of these curves represents the changing likelihood of a given response from birth to 72 months of age. Multiplying the curves then offers an estimate of the likelihood of observing the entire pattern of responses reported at each age from birth to 72 months. The maximum value of this curve represents the age at which the pattern of responses is most likely to be observed, and we use this maximum value as our estimate of each child's developmental age. For children who received the maximum score (all answers were "2: very much"), developmental age was set just above the maximum attainable developmental age resulting from at least 1 imperfect response on the set of questions their parents completed. Developmental age for children who received minimum scores (all answers were "0: not at all") was set in a similar fashion. An advantage of the item response curve method is that developmental status is estimated on the basis of all available milestones, even if not every question is answered. To achieve model convergence, responses for 2 items were recoded as binary by combining "not at all" and "somewhat" responses.

Set up and Logistics

```
rm(list = ls())
library(ggplot2)
library(tidyverse)
library(stringr)
library(knitr)
library(kableExtra)
library(ggpubr)
library(data.table)
library(dtplyr)
```

Authenticate Lab in SQL Big Query

```
library(bigrquery)
```

Loading required package: bigrquery

```
bq_auth()  
project_id <- if (exists('project_id')) project_id else bq_projects()[1]
```

There are certain time-consuming aspects of the pipeline that you may not want to rerun. If you have already run the pipeline before, you can set `restart = FALSE` to reuse files.

```
restart <- TRUE
```

Download and Clean SWYC Codes

Query SWYC forms

First, let's extract `qform` data that contains `swyc` or `developmental milestones`. This query joins the questionnaire answers (`qform_answer`) with the `encounter` and `patient` tables to retrieve the necessary demographic and visit-level data (like gestational age for prematurity adjustments).

```
dir.create("queries", showWarnings = F)  
file <- "queries/qforms_developmental_milestones.csv"  
if (file.exists(file) & !restart) {  
  qforms.swyc <- read_csv(file, show_col_types = FALSE)  
} else {  
  # Query the centralized clinical data warehouse  
  table_query <- "SELECT  
                  enc.pat_id,  
                  pat.sex_abbr,  
                  pat.race,  
                  pat.ethnicity,  
                  pat.gestational_age_num,  
                  enc.encounter_id,  
                  enc.effective_age,  
                  qform.form_name,  
                  qform.form_friendly_name,
```

```

        qform.form_section_name,
        qform.form_section_friendly_name,
        qform.form_line,
        qform.quest_type_name,
        qform.quest_name,
        qform.quest_text,
        qform.answer_line,
        qform.numeric_answer,
        qform.text_answer,
        enc.enc_type_name,
        enc.appt_status_code,
        enc.appt_status_name,
        enc.appt_prc_name,
        pat.interpreter_needed,
        pat.birth_weight_kg
    FROM arcus.qform_answer qform
    INNER JOIN arcus.encounter enc on qform.encounter_id = enc.encounter_id
    INNER JOIN arcus.patient pat on qform.pat_id=pat.pat_id
    WHERE lower(form_friendly_name) LIKE '%developmental milestones%' OR
           lower(form_friendly_name) LIKE '%swyc%'

qforms.swyc <- bq_table_download(bq_project_query(project_id, table_query))

write_csv(qforms.swyc,file)
}

```

While the clinical interpretation of the SWYC is traditionally based on exact chronological ages calculated using the specific month and day of birth, this is not possible here due to the de-identification process (removal of exact datetimes) in certain Arcus labs. Therefore, we calculate chronological age based on the **effective_age** variable (age in days).

The SWYC manual also requires prematurity adjustments when the child is less than 24 months old AND born 3 or more weeks premature. We calculate **clin.age** to reflect this adjustment for strict clinical scoring. Note that these prematurity adjustments will *not* be used for the continuous normative scores later, as Sheldrick et al (2019) excluded children born before 37 weeks gestation. I have elected to retain these samples for completeness, and so it is up to the researcher to decide whether to exclude them.

```

# Convert effective age in days to months
qforms.swyc$norm.age <- qforms.swyc$effective_age/30.437

```

```

# Check for prematurity (Standard full term is 40 weeks)
qforms.swyc$weeks_premature <- 40 - as.numeric(qforms.swyc$gestational_age_num)

# Adjust for prematurity if swyc age is less than 24 months
qforms.swyc <- qforms.swyc %>%
  mutate(clin.age = case_when(
    is.na(weeks_premature) ~ norm.age,
    # If >= 3 weeks early AND under 24 months, subtract the prematurity
    weeks_premature >= 3 & norm.age < 24 ~ norm.age - weeks_premature,
    .default = norm.age)) %>%
  mutate(norm.age = floor(norm.age),
         clin.age = floor(clin.age),)

qforms.swyc %>%
  mutate(born_premature = weeks_premature >= 3) %>%
  mutate(born_premature = case_when(born_premature == T ~ "Premature",
                                   born_premature == F ~ "Non-Premature",
                                   .default = "Unknown")) %>%
  group_by(born_premature) %>% summarize(N = n()) %>% kable()

```

born_premature	N
Non-Premature	217614
Premature	106621
Unknown	20019

Standardize SWYC Names

Questionnaire names in the EHR are often inconsistently formatted. Here, we standardize the strings so that all individuals using the same age-targeted questionnaire are appropriately grouped.

```

library(stringr)
qforms.swyc <- qforms.swyc %>%
  # Remove trailing age brackets and standardize 'month' casing
  mutate(form_friendly_name = str_replace_all(form_friendly_name, " \\(\\d*\\)", "")) %>%
  mutate(form_friendly_name = str_replace_all(form_friendly_name, "month", "Month")) %>%
  mutate(form_friendly_name = str_replace_all(form_friendly_name, "Months", "Month")) %>%
  mutate(form_friendly_name = str_replace_all(form_friendly_name, " ", "")) %>%

  # Standardize the prefixes

```

```

mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Survey of Well-being of Young Children: ",
      "SWYC ")) %>%

# Map specific EHR form names to the standardized SWYC target age buckets
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(4 Month\\)",
      "Developmental Milestones: 4 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(6|7|8) Month\\)",
      "Developmental Milestones: 6 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(9|10|11) Month\\)",
      "Developmental Milestones: 9 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(12|13|14) Month\\)",
      "Developmental Milestones: 12 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(15|16) Month\\)",
      "Developmental Milestones: 15 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(17|18|19|22) Month\\)",
      "Developmental Milestones: 18 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(24|26) Month\\)",
      "Developmental Milestones: 24 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(29|30|34) Month\\)",
      "Developmental Milestones: 30 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(35|36|46) Month\\)",
      "Developmental Milestones: 36 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(48|49) Month\\)",
      "Developmental Milestones: 48 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones \\\(60 Month\\)",
      "Developmental Milestones: 60 Month")) %>%

# Final cleanup for zero-padding single digits to ensure proper sorting later
mutate(form_friendly_name = str_replace_all(form_friendly_name,
      "Developmental Milestones", "")) %>%

```

```

mutate(form_friendly_name = str_replace_all(form_friendly_name,
                                             ": 4 Month", ": 04 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
                                             ": 6 Month", ": 06 Month")) %>%
mutate(form_friendly_name = str_replace_all(form_friendly_name,
                                             ": 9 Month", ": 09 Month"))

# Print counts of each SWYC assessment
qforms.swyc.counts <- qforms.swyc %>%
  dplyr::select(c(pat_id, form_friendly_name)) %>%
  distinct() %>%
  count(form_friendly_name)

kable(qforms.swyc.counts)

```

form_friendly_name	n
SWYC : 04 Month	12
SWYC : 06 Month	694
SWYC : 09 Month	6155
SWYC : 12 Month	54
SWYC : 15 Month	1445
SWYC : 18 Month	5915
SWYC : 24 Month	7238
SWYC : 30 Month	3975
SWYC : 36 Month	31
SWYC : 48 Month	25
SWYC : 60 Month	29
SWYC Baby Pediatric Symptoms Checklist (BPSC)	25
SWYC Family Questions	59
SWYC Preschool Pediatric Symptom List (PPSC)	14

Filter Questionnaires

Let's restrict our dataset to just the developmental milestone forms (excluding behavioral/family risk portions of the SWYC).

```
qforms.swyc <- qforms.swyc[grepl("Month", qforms.swyc$form_friendly_name),]
```

We could also exclude low-N forms, say < 100 . However, since we're generating continuous normative scores anyway, maintaining these edge cases is valuable for longitudinal trajectories.

```
# qforms.swyc <- qforms.swyc[qforms.swyc$form_friendly_name %in%
#                               qforms.swyc.counts$form_friendly_name[qforms.swyc.counts$n >= 1]

# Print counts of each SWYC assessment
qforms.swyc.counts <- qforms.swyc %>%
  dplyr::select(c(pat_id, form_friendly_name)) %>%
  distinct() %>%
  count(form_friendly_name) %>%
  select(`SWYC Form` = form_friendly_name,
         Count = n)

kable(qforms.swyc.counts)
```

SWYC Form	Count
SWYC : 04 Month	12
SWYC : 06 Month	694
SWYC : 09 Month	6155
SWYC : 12 Month	54
SWYC : 15 Month	1445
SWYC : 18 Month	5915
SWYC : 24 Month	7238
SWYC : 30 Month	3975
SWYC : 36 Month	31
SWYC : 48 Month	25
SWYC : 60 Month	29

```
rm(qforms.swyc.counts)
```

Now let's check to see if a given form was clinically appropriate based on the child's adjusted age range at the time of the encounter.

```
# Import SWYC score threshold, which contain the age at which each SWYC should be administered
swyc_grader <- read_csv("data/SWYC_norms/SWYC_scores.csv", show_col_types = FALSE)

# Join the official thresholds and forms to the actual data
qforms.swyc <- left_join(qforms.swyc, swyc_grader, by = c("clin.age" = "Age"))
rm(swyc_grader)
# Check for individuals who may have received the wrong form.
qforms.swyc %>%
  mutate(correct_swyc = form_friendly_name == Form) %>%
```



```
group_by(correct_swyc) %>%
  summarize(N = n()) %>%
  kable()
```

correct_swyc	N
FALSE	16989
TRUE	325361

In SCIT-605, about 17,000 entries (5%) don't match the expected form (e.g., a 24-month form given to a 15-month-old). These could represent the calculated age (using `effective_age`) not reflecting the actual age in months. However, they could also represent clinical errors or off-label use in the EHR. The vast majority are one age-group off (e.g., 6 month vs 9 month). We will remove them to ensure validity.

```
qforms.swyc <- qforms.swyc[qforms.swyc$form_friendly_name == qforms.swyc$Form,]
```

Let's collect the scoring question for all SWYCs. These will serve as our official score sums for use in calculating clinical threshold.

```
qforms.swyc.scores <- qforms.swyc[grepl("Scoring Question",qforms.swyc$quest_type_name),]
qforms.swyc.scores$cat <- "Score"
qforms.swyc.scores <- qforms.swyc.scores[order(qforms.swyc.scores$effective_age),]
qforms.swyc.scores$pat_id <- factor(qforms.swyc.scores$pat_id, levels = unique(qforms.swyc.s
```

Visualize SWYC

Let's add some helpful visualizations of the density of SWYC scores and longitudinal SWYCs.

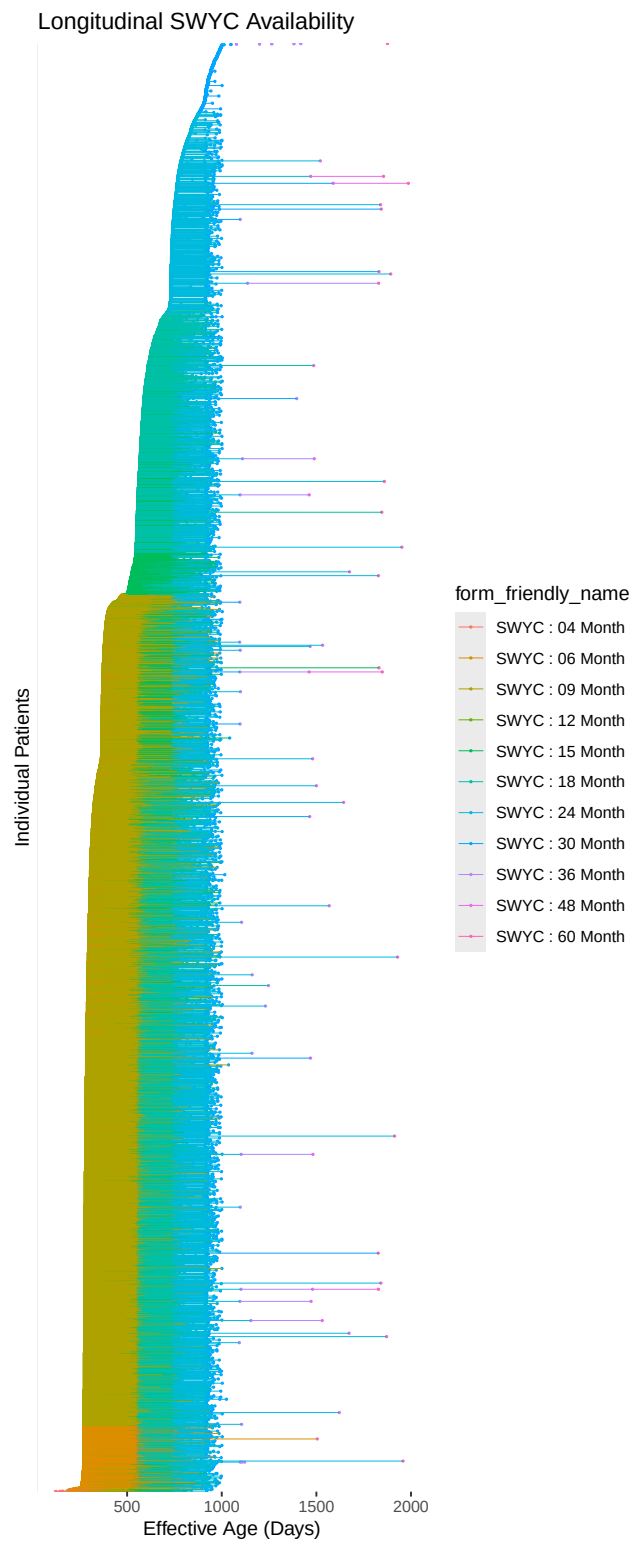
Plot the Distribution of Scores for Each SWYC

Plot the longitudinal assessments for each patient

```
# Visualize the longitudinal density of SWYC completion across the cohort
ggplot(qforms.swyc.scores, aes(x = effective_age, y = pat_id, color = form_friendly_name)) +
  geom_point(size = 0.2) +
  geom_line(aes(group = pat_id), size = 0.1) +
```

```
theme(axis.text.y = element_blank(),  
      axis.ticks.y = element_blank()) +  
labs(x = "Effective Age (Days)", y = "Individual Patients", title = "Longitudinal SWYC Awa.
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



```
dir.create("results/SWYC/", showWarnings = F)
ggsave("results/SWYC/swyc_ages.png")
```

Saving 5 x 12 in image

Estimate SWYC Severity Approach 1 (Clinical Thresholds)

The SWYC has specific categorical guidance on whether individuals are meeting developmental milestones based on their age and total response sum. Let's apply those strict clinical thresholds to this data to flag "at-risk" individuals.

```
# We've already added the clinical thresholds to each individual, so now let's apply the thresholds
# Create a Boolean flag if the raw score exceeds the established clinical cut score
qforms.swyc.scores$clin.meets_milestones <- qforms.swyc.scores$numeric_answer > qforms.swyc.scores$clin_cut_score

# And let's visualize how many individuals are meeting clinical milestone thresholds
qforms.swyc.scores %>%
  group_by(form_friendly_name) %>%
  summarize(`Met Milestones` = sum(clin.meets_milestones == T, na.rm = T),
            `Not Met Milestones` = sum(clin.meets_milestones == F, na.rm = T),
            `Prop Met Milestones` = round(`Met Milestones`/n(),2)) %>%
  kable(caption = "Meeting Developmental Milestones")
```

Table 5: Meeting Developmental Milestones

form_friendly_name	Met Milestones	Not Met Milestones	Prop Met Milestones
SWYC : 04 Month	5	3	0.62
SWYC : 06 Month	397	246	0.62
SWYC : 09 Month	4518	1233	0.78
SWYC : 12 Month	20	16	0.56
SWYC : 15 Month	1144	296	0.79
SWYC : 18 Month	1795	3751	0.32
SWYC : 24 Month	6073	1332	0.81
SWYC : 30 Month	2922	1120	0.72
SWYC : 36 Month	25	6	0.81
SWYC : 48 Month	15	8	0.65
SWYC : 60 Month	15	6	0.71

Estimate SWYC Severity Approach 1 (Normative Scores)

I've discussed this approach from Sheldrick et al., (2019) at the top of the notebook. Let's apply it here. The approach takes any combination of questions answered by an individual and calculates the exact age at which that specific combination of responses is most likely to occur. This gives us the **developmental age**. We can then divide **developmental age** by the actual chronological age to generate a continuous **developmental quotient** (DQ).

Get Response Rate

First, let's plot the number of responses we have for each question. The IRT technique is highly robust against missing data, but assessing response bias is still crucial.

```
qforms.swyc.response <- table(qforms.swyc$form_friendly_name,qforms.swyc$form_line)

response.rate <- qforms.swyc %>%
  arrange(form_line) %>%
  dplyr::select(form_friendly_name, form_line) %>%
  pivot_wider(id_cols = "form_friendly_name",
              names_from = "form_line",
              values_from = "form_line",
              values_fn = function(x) sum(!is.na(x)), values_fill = 0)

kable(response.rate, caption = "Responses by SWYC")
```

Table 6: Responses by SWYC

form_friendly_name	2	3	4	5	6	7	8	9	10	11	12	13	
SWYC : 30 Month	4089	4077	4069	4066	4061	4057	4057	4056	4055	4051	4049	4047	4042
SWYC : 36 Month	31	31	31	31	31	31	31	31	31	31	31	31	30
SWYC : 48 Month	23	23	23	23	23	23	23	23	23	23	23	23	23
SWYC : 60 Month	22	21	21	21	21	21	21	21	21	21	21	21	21
SWYC : 12 Month	36	36	36	36	36	36	36	36	36	36	36	36	36
SWYC : 06 Month	646	643	643	643	643	643	643	643	643	643	643	643	641

form_friendly_name	2	3	4	5	6	7	8	9	10	11	12	13	
SWYC : 09 Month	5814	5805	5782	5776	5769	5766	5763	5761	5757	5757	5754	5753	5752
SWYC : 18 Month	5630	5600	5571	5560	5557	5556	5555	5552	5551	5551	5550	5549	5544
SWYC : 15 Month	1460	1452	1446	1444	1444	1442	1442	1441	1441	1441	1441	1441	1440
SWYC : 04 Month	8	8	8	8	8	8	8	8	8	8	8	8	8
SWYC : 24 Month	7509	7479	7443	7434	7424	7419	7416	7410	7409	7409	7407	7406	7403

```
rm(response.rate)
```

We tend to have a slight drop-off in the later questions on the forms, indicating survey fatigue, but nothing severe enough to bias the IRT estimates.

Get List of Questions

Now let's extract the individual questions. Many EHR question texts have subtle typos or formatting artifacts that must be cleaned via regex to match the exact string keys in the Sheldrick normative parameters file.

```
qforms.swyc.questions <- qforms.swyc %>%
  dplyr::select(c(pat_id, encounter_id, form_friendly_name,
                 form_line, quest_type_name, quest_text, form_section_friendly_name,
                 numeric_answer, norm.age)) %>%
  # Standardize 'Scoring' identifiers
  mutate(quest_text = ifelse(grepl("SCORING QUESTION",toupper(quest_text)),
                             "Scoring Question", quest_text),
         quest_text = ifelse(grepl("MONTH SCORE",toupper(quest_text)),
                             "Scoring Question", quest_text),
         quest_text = ifelse(grepl("TOTAL DEVELOPMENT SCORE",toupper(quest_text)),
                             "Scoring Question", quest_text)) %>%
  # Clean EHR text artifacts to perfectly match Sheldrick strings
  mutate(quest_text = str_replace_all(quest_text, 'ga, " ', 'ga, '),
         quest_text = str_replace_all(quest_text, 'ma, "', 'ma, '),
         quest_text = str_replace_all(quest_text, '"', '"'),
         quest_text = str_replace_all(quest_text, "...", "..."),
         quest_text = str_replace_all(quest_text, 'blanket\\(?:$|', 'blanket?"),
```

```

      quest_text = trimws(quest_text)) %>%
    filter(!(quest_text %in% c("Respondent"))) %>%
    filter(!grepl("PLEASE BE SURE",toupper(quest_text))) %>%
    drop_na(numeric_answer)

write_csv(qforms.swyc.questions, "results/SWYC/SWYC_questions.csv")

indiv QUESTS <- qforms.swyc.questions %>%
  dplyr::select(quest_text) %>%
  count(quest_text)

kable(indiv QUESTS, caption = "Number of Responses per Question")

```

Table 7: Number of Responses per Question

quest_text	n
Answers questions like “What do you do when you are cold?” or “...when you are sleepy?”	4096
Asks questions beginning with “why” or “how” - like “Why no cookie?”	4082
Brings hands together	8
Calls you “mama” or “dada” or similar name	7238
Climbs up a ladder at a playground	12971
Compares things - using words like “bigger” or “shorter”	4101
Copies sounds that you make	7233
Draws lines	11459
Draws pictures you recognize	44
Draws simple shapes - like a circle or a square	75
Explains the reasons for things, like needing a sweater when it’s cold	4079
Follows directions - like “Come here” or “Give me the ball”	7229
Follows simple rules when playing a board game or card game	44
Gets to a sitting position by him or herself	6411
Holds head steady when being pulled up to a sitting position	8
Holds two objects and bangs them together	651
Holds up arms to be picked up	6414
Jumps off the ground with two feet	12963
Keeps head steady when held in a sitting position	8
Kicks a ball	6995
Laughs	8
Looks around when you say things like “Where’s your bottle?” or “Where’s your blanket?”	7236
Looks for you or another caregiver when upset	651

quest_text	n
Looks when you call his or her name	651
Makes sounds like “ga”, “ma”, or “ba”	651
Names at least 5 body parts - like nose, hand, or tummy	14415
Names at least 5 familiar objects - like ball or milk	6992
Names at least one color	11472
Names the days of the week in the correct order	21
Passes a toy from one hand to the other	651
Picks up food and eats it	6445
Plays games like “peek-a-boo” or “pat-a-cake”	5797
Prints his or her name	44
Pulls up to standing	6440
Puts 2 or more words together - like “more water” or “go outside”	12956
Rolls over	651
Runs	7035
Says his or her first name when asked	11461
Says words like “feet” for more than one foot and “men” for more than one man	75
Scoring Question	24946
Stays dry all night	44
Stays in the lines when coloring	21
Talks so other people can understand him or her most of the time	4087
Tells you a story from a book or tv	75
Tries to get you to watch by saying “Look at me”	11466
Uses words like “me” or “mine”	12967
Uses words like “yesterday” and “tomorrow” correctly	75
Uses words to ask for help	12952
Walks across a room without help	7231
Walks up stairs with help	7033
Washes and dries hands without help (even if you turn on the water)	4086

```
rm(indiv_quests)
```

Calculate Developmental Quotient

Requisite File: Sheldrick_et_al_2019_norms.csv

This file contains normative parameters (α_1 , α_2 , and β) for the response curves. The parameters were manually extracted from the Excel supplement file in the paper. Questions found only in the two month assessment were not extracted, as this assessment is considered to be less reliable.


```
# Load Sheldrick norms
sheldrick_norms <- read_csv("data/SWYC_norms/Sheldrick_et_al_2019_norms.csv", show_col_types = FALSE)
sheldrick_norms <- sheldrick_norms[!is.na(sheldrick_norms$`SWYC Milestone Text`),]

# Merge the IRT parameters onto the patient responses based on the question text
qforms.swyc.questions <- merge(qforms.swyc.questions, sheldrick_norms,
                               by.x = "quest_text", by.y = "SWYC Milestone Text")
rm(sheldrick_norms)

# Recode responses from EHR format (1-3) to Sheldrick format (0 = Not Yet, 1 = Somewhat, 2 = Very)
qforms.swyc.questions$numeric_answer <- qforms.swyc.questions$numeric_answer - 1
```

To calculate the developmental quotient (DQ) from Sheldrick et al. (2019), we use a Graded Response Model (a type of IRT). For every single patient response, we calculate the probability of that specific answer occurring at *every* theoretical age from 0 to 72 months using the logistic equations.

```
for (mo in 1:72) {
  log_mo <- log(mo)

  # Generate the logistic response curve for each question based on the child's answer
  # (0, 1, or 2)
  qforms.swyc.questions <- qforms.swyc.questions %>%
    mutate("prob.{mo}" := case_match(numeric_answer,
                                       # P(Score = 0)
                                       0 ~ 1 - (exp(-alpha1 + beta*log_mo) /
                                                (1+exp(-alpha1 + beta*log_mo))),
                                       # P(Score = 1)
                                       1 ~ (1-exp(-alpha2 + beta*log_mo) /
                                                (1+exp(-alpha2 + beta*log_mo)) -
                                                (1-(exp(-alpha1 + beta*log_mo) /
                                                (1+exp(-alpha1 + beta*log_mo))))) ,
                                       # P(Score = 2)
                                       2 ~ exp(-alpha2 + beta*log_mo) /
                                                (1+exp(-alpha2 + beta*log_mo)),
                                       ))
}
```

Now we compute Maximum Likelihood Estimation. We multiply the probabilities of all responses given by the patient at each theoretical age (0-72). The age where the combined probability peaks is designated as the **developmental age**.

Edge Case Handling: This mathematical method approaches infinity/zero when responses are entirely perfect (all 2s) or completely absent (all 0s). Per the Sheldrick supplement, we simulate a single imperfect response to find the ceiling/floor ages for these children. This step can be quite time intensive.

```
calc_devo_age <- function(df) {
  setDT(df)

  # 1. Calculate the product of each column starting with "prob."
  col_prods <- df[, lapply(.SD, prod), .SDcols = patterns("^prob\\.([0-9]+)$")]

  # 2. Convert the resulting 1-row data.table to a standard numeric vector
  prod_vec <- as.numeric(col_prods)

  # 3. Find the maximum probability and its index (which corresponds to the age)
  best_age <- which.max(prod_vec)

  # 4. Create and return the simulated devo age
  return(list(devo.prob = prod_vec[best_age], devo.age = best_age))
}

sim_imperfect_response <- function(j, df) {
  # Calculate new curve assuming the jth response is 1 (Somewhat) to provide an anchor
  for (mo in 1:72) {
    log_mo <- log(mo)
    df[j,] <- df[j,] %>%
      mutate("prob.{mo}" := (1-exp(-alpha2 + beta*log_mo) /
                             (1+exp(-alpha2 + beta*log_mo)) -
                             (1-(exp(-alpha1 + beta*log_mo) /
                                (1+exp(-alpha1 + beta*log_mo)))))
  }
  return(calc_devo_age(df))
}

rm(response_summary, qforms.swyc)
gc()

library(parallel)

if (file.exists("results/SWYC/individual_development_scores.csv") & !restart) {
  qforms.swyc.indiv <- read_csv("results/SWYC/individual_development_scores.csv",
```

```

                                show_col_types = FALSE)
} else {
  # Get data table of subjects
  qforms.swyc.indiv <- lazy_dt(qforms.swyc.questions) %>%
    dplyr::select(c(pat_id, encounter_id, form_friendly_name, norm.age))
    mutate(devo.age = NA,
           devo.prob = NA) %>%
    distinct() %>%
    as.data.table()

  # Reduce data table of question responses for efficiency
  qforms.swyc.questions <- lazy_dt(qforms.swyc.questions) %>%
    select(encounter_id, form_friendly_name, norm.age, numeric_
           beta, alpha1, alpha2, starts_with("prob.")) %>%
    drop_na() %>%
    as.data.table()

  # Record QA data and check for boundary_conditions
  response_summary <- lazy_dt(qforms.swyc.questions) %>%
    dplyr::select(c(encounter_id, form_friendly_name, norm.age, numeric_
    na.omit() %>%
    group_by(encounter_id, form_friendly_name, norm.age) %>%
    summarize(n_answers = n(),
              all_0s = all(numeric_answer == 0),
              all_2s = all(numeric_answer == 2),
              sum_answers = sum(numeric_answer)) %>%
    as.data.table()

  qforms.swyc.indiv <- lazy_dt(qforms.swyc.indiv) %>%
    left_join(response_summary) %>%
    as.data.table()

  # Loop through each clinical encounter and find the maximum of the probability curve
  it_max <- nrow(qforms.swyc.indiv)

  pb = txtProgressBar(min = 0, max = it_max, initial = 0, style = 3)

  for (i in 1:it_max) {
    # Subset data to include only rows that belong to the current patient visit
    qforms.ind <- lazy_dt(qforms.swyc.questions) %>%
      filter(encounter_id == qforms.swyc.indiv$encounter_id[i] &

```

```

        form_friendly_name == qforms.swyc.indiv$form_friendly_name[i] &
        norm.age == qforms.swyc.indiv$norm.age[i]) %>%
select(beta, alpha1, alpha2, starts_with("prob.")) %>%
as.data.table()

# Skip encounters with insufficient data to estimate a curve
if(nrow(qforms.ind) < 5) {
  next
# Resolve duplicates: take the form with the highest completion rate
} else if (length(qforms.ind$form_section_friendly_name) > 1) {
  qforms.ind <- qforms.ind %>%
    filter(form_section_friendly_name ==
           names(which.max(table(form_section_friendly_name))))
}

# Standard Maximum Likelihood Estimation
if (!(qforms.swyc.indiv$all_0s[i] | qforms.swyc.indiv$all_2s[i])) {
  # Calculate the most probable developmental age
  res <- calc_devo_age(qforms.ind)

  # Save the devo age
  qforms.swyc.indiv$devo.age[i] <- res$devo.age
  qforms.swyc.indiv$devo.prob[i] <- res$devo.prob
} else {
  # Boundary Condition Execution: Simulate imperfect responses using parallel processing
  sim_responses <- mcmapply(sim_imperfect_response, 1:nrow(qforms.ind), MoreArgs = list(
    mc.cores = parallelly::availableCores())
  # Apply age adjustment to place the developmental age just outside the realm of possible
  if (qforms.swyc.indiv$all_0s[i]) {
    # Set devo.age to the lowest possible age - 1
    min.ind <- which.min(unlist(sim_responses["devo.age",]))
    qforms.swyc.indiv$devo.age[i] <- sim_responses["devo.age",min.ind]$devo.age - 1
    qforms.swyc.indiv$devo.prob[i] <- sim_responses["devo.prob",min.ind]$devo.prob
  } else if (qforms.swyc.indiv$all_2s[i]) {
    # Set devo.age to the highest possible age + 1
    max.ind <- which.max(unlist(sim_responses["devo.age",]))
    qforms.swyc.indiv$devo.age[i] <- sim_responses["devo.age",max.ind]$devo.age + 1
    qforms.swyc.indiv$devo.prob[i] <- sim_responses["devo.prob",max.ind]$devo.prob
  }
}
}

```

```

    setTxtProgressBar(pb,i)
  }

# Calculate final Developmental Quotient (DQ)
qforms.swyc.indiv <- qforms.swyc.indiv %>%
  mutate(devo.quotient = devo.age/norm.age) %>%
  as_tibble()

write_csv(qforms.swyc.indiv,"results/SWYC/individual_development_scores.csv")
}

```

Formatting

Now we can merge these calculated continuous scores with our clinical flag metrics and meta-data.

```

qforms.swyc.indiv <- tibble(merge(qforms.swyc.indiv, qforms.swyc.scores,
                                by = c("pat_id","encounter_id","form_friendly_name",
                                      "norm.age"))) %>%
  mutate(well_visit = grepl("well",tolower(appt_prc_name)),
         clin.matches_expected_swyc = form_friendly_name == Form)

```

Clean up the resulting dataframe by organizing the final columns and dropping unneeded temporary variables.

```

qforms.swyc.indiv <- qforms.swyc.indiv %>%
  select_if(~sum(!is.na(.)) > 0) %>%
  relocate(
    sex_abbr, race, ethnicity,
    well_visit, effective_age,
    age_mo = norm.age,
    gestational_age_num,
    weeks_premature,
    SWYC.form_friendly_name = form_friendly_name,
    SWYC.n_answers = n_answers,
    SWYC.sum_answers = sum_answers,
    SWYC.count.score = numeric_answer,
    SWYC.all_0s = all_0s, SWYC.all_2s = all_2s,
    clin.prematurity_adj_age = clin.age,
    clin.meets_milestones, clin.matches_expected_swyc,
    devo.age, devo.quotient, devo.prob, appt_prc_name,

```

```

    birth_weight_kg, .after = encounter_id) %>%
dplyr::select(c(pat_id:birth_weight_kg)) %>%
# Ensure validity: must have >=5 answers and match the age form
filter(SWYC.n_answers >= 5,
       clin.matches_expected_swyc)

```

Final Visualizations

Visualize Developmental Quotient Distribution

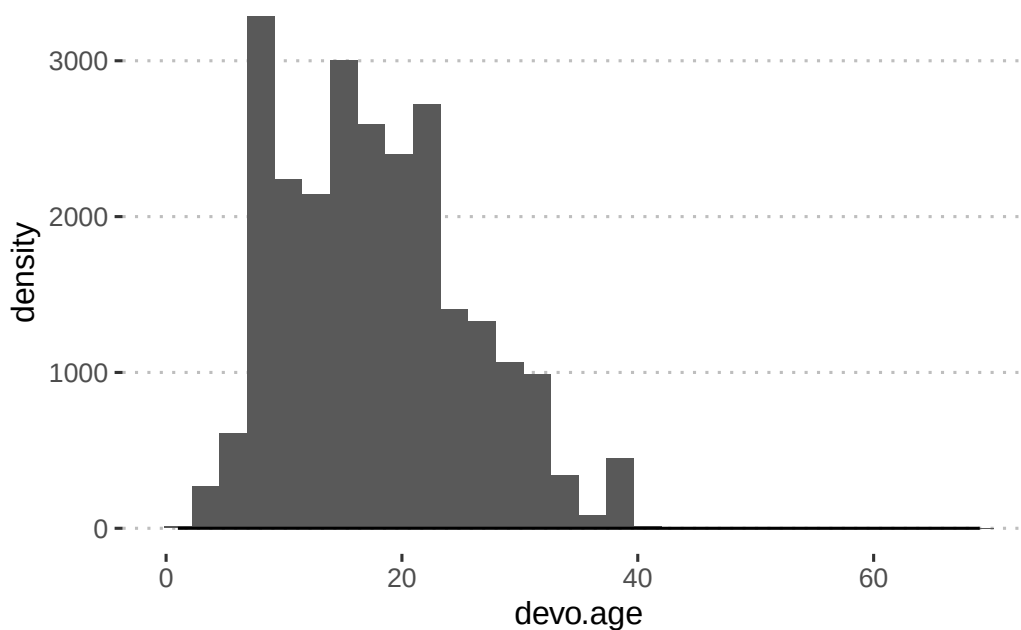
Let's plot the distributions to visually verify our models. Ideally, developmental quotient should be relatively normally distributed around 1.0.

```

ggplot(qforms.swyc.indiv, aes(x = devo.age)) +
  geom_histogram() +
  geom_density() +
  theme_pubclean()

```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
labs(title = "Distribution of Computed Developmental Age")
```

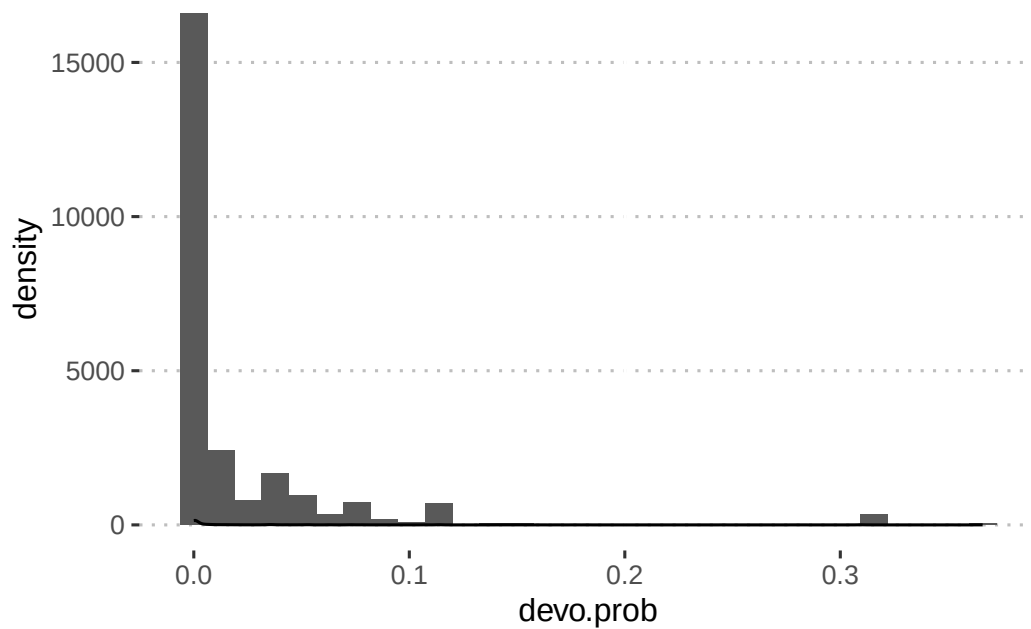
```
<ggplot2::labels> List of 1  
 $ title: chr "Distribution of Computed Developmental Age"
```

```
ggsave("results/SWYC/development_age.png")
```

Saving 5.5 x 3.5 in image
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

```
ggplot(qforms.swyc.indiv, aes(x = devo.prob)) +  
  geom_histogram() +  
  geom_density() +  
  theme_pubclean()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



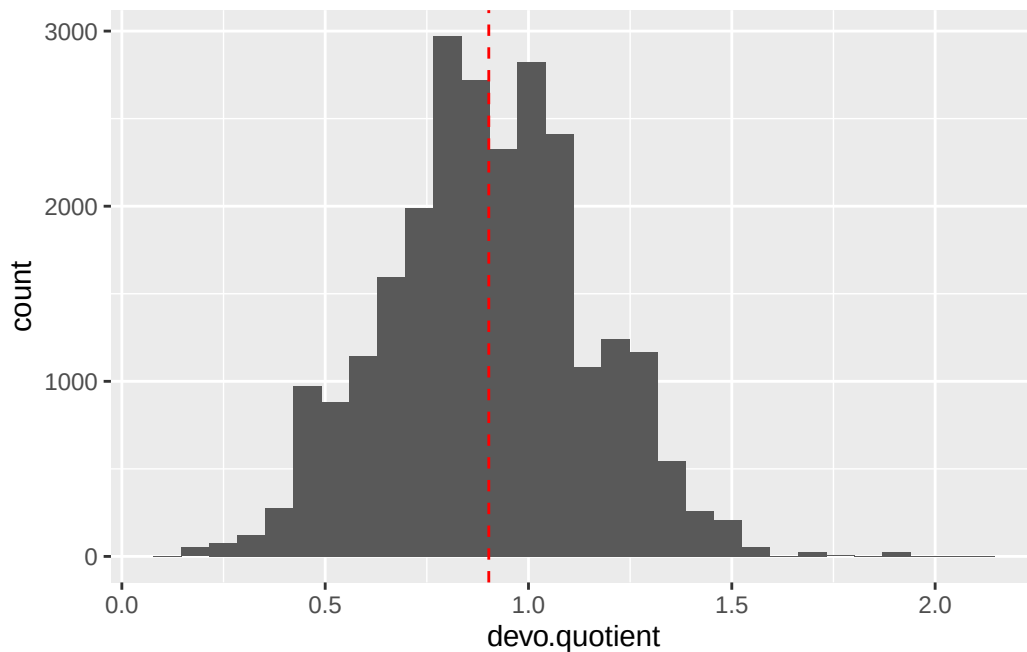
```
ggsave("results/SWYC/development_prob.png")
```

Saving 5.5 x 3.5 in image

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

```
ggplot(qforms.swyc.indiv, aes(x = devo.quotient)) +  
  geom_histogram() +  
  geom_vline(xintercept = mean(qforms.swyc.indiv$devo.quotient),  
            linetype = "dashed", color = "red")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
theme_pubclean() +  
labs(title = "Distribution of Developmental Quotient (DQ)")
```

<theme> List of 144

```
$ line : <ggplot2::element_line>  
..@ colour      : chr "black"  
..@ linewidth   : num 0.545  
..@ linetype    : num 1  
..@ lineend     : chr "butt"  
..@ linejoin    : chr "round"  
..@ arrow       : logi FALSE
```



```

..@ arrow.fill      : chr "black"
..@ inherit.blank: logi TRUE
$ rect                                     : <ggplot2::element_rect>
..@ fill            : chr "white"
..@ colour          : chr "black"
..@ linewidth       : num 0.545
..@ linetype        : num 1
..@ linejoin        : chr "round"
..@ inherit.blank: logi TRUE
$ text                                     : <ggplot2::element_text>
..@ family          : chr ""
..@ face            : chr "plain"
..@ italic          : chr NA
..@ fontweight      : num NA
..@ fontwidth       : num NA
..@ colour          : chr "black"
..@ size            : num 12
..@ hjust           : num 0.5
..@ vjust           : num 0.5
..@ angle           : num 0
..@ lineheight      : num 0.9
..@ margin          : <ggplot2::margin> num [1:4] 0 0 0 0
..@ debug           : logi FALSE
..@ inherit.blank: logi TRUE
$ title                                     : chr "Distribution of Developmental Quotient (DQ)"
$ point                                     : <ggplot2::element_point>
..@ colour          : chr "black"
..@ shape           : num 19
..@ size            : num 1.64
..@ fill            : chr "white"
..@ stroke          : num 0.545
..@ inherit.blank: logi TRUE
$ polygon                                     : <ggplot2::element_polygon>
..@ fill            : chr "white"
..@ colour          : chr "black"
..@ linewidth       : num 0.545
..@ linetype        : num 1
..@ linejoin        : chr "round"
..@ inherit.blank: logi TRUE
$ geom                                     : <ggplot2::element_geom>
..@ ink             : chr "black"
..@ paper           : chr "white"
..@ accent          : chr "#3366FF"

```

```

..@ linewidth   : num 0.545
..@ borderwidth: num 0.545
..@ linetype    : int 1
..@ bordertype  : int 1
..@ family      : chr ""
..@ fontsize    : num 4.22
..@ pointsize   : num 1.64
..@ pointshape  : num 19
..@ colour      : NULL
..@ fill        : NULL
$ spacing                               : 'simpleUnit' num 6points
  ..- attr(*, "unit")= int 8
$ margins                               : <ggplot2::margin> num [1:4] 6 6 6 6
$ aspect.ratio                          : NULL
$ axis.title                            : NULL
$ axis.title.x                          : <ggplot2::element_text>
  ..@ family      : NULL
  ..@ face        : NULL
  ..@ italic      : chr NA
  ..@ fontweight   : num NA
  ..@ fontwidth    : num NA
  ..@ colour      : NULL
  ..@ size        : NULL
  ..@ hjust       : NULL
  ..@ vjust       : num 1
  ..@ angle       : NULL
  ..@ lineheight   : NULL
  ..@ margin      : <ggplot2::margin> num [1:4] 3 0 0 0
  ..@ debug       : NULL
  ..@ inherit.blank: logi TRUE
$ axis.title.x.top                      : <ggplot2::element_text>
  ..@ family      : NULL
  ..@ face        : NULL
  ..@ italic      : chr NA
  ..@ fontweight   : num NA
  ..@ fontwidth    : num NA
  ..@ colour      : NULL
  ..@ size        : NULL
  ..@ hjust       : NULL
  ..@ vjust       : num 0
  ..@ angle       : NULL
  ..@ lineheight   : NULL
  ..@ margin      : <ggplot2::margin> num [1:4] 0 0 3 0

```

```

..@ debug          : NULL
..@ inherit.blank: logi TRUE
$ axis.title.x.bottom      : NULL
$ axis.title.y          : <ggplot2::element_text>
..@ family          : NULL
..@ face            : NULL
..@ italic          : chr NA
..@ fontweight      : num NA
..@ fontwidth       : num NA
..@ colour          : NULL
..@ size            : NULL
..@ hjust           : NULL
..@ vjust           : num 1
..@ angle           : num 90
..@ lineheight      : NULL
..@ margin          : <ggplot2::margin> num [1:4] 0 3 0 0
..@ debug          : NULL
..@ inherit.blank: logi TRUE
$ axis.title.y.left       : NULL
$ axis.title.y.right      : <ggplot2::element_text>
..@ family          : NULL
..@ face            : NULL
..@ italic          : chr NA
..@ fontweight      : num NA
..@ fontwidth       : num NA
..@ colour          : NULL
..@ size            : NULL
..@ hjust           : NULL
..@ vjust           : num 1
..@ angle           : num -90
..@ lineheight      : NULL
..@ margin          : <ggplot2::margin> num [1:4] 0 0 0 3
..@ debug          : NULL
..@ inherit.blank: logi TRUE
$ axis.text              : <ggplot2::element_text>
..@ family          : NULL
..@ face            : NULL
..@ italic          : chr NA
..@ fontweight      : num NA
..@ fontwidth       : num NA
..@ colour          : chr "#4D4D4DFF"
..@ size            : 'rel' num 0.8
..@ hjust           : NULL

```

```

..@ vjust      : NULL
..@ angle      : NULL
..@ lineheight  : NULL
..@ margin     : NULL
..@ debug      : NULL
..@ inherit.blank: logi TRUE
$ axis.text.x      : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA
..@ colour      : NULL
..@ size        : NULL
..@ hjust       : NULL
..@ vjust       : num 1
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : <ggplot2::margin> num [1:4] 2.4 0 0 0
..@ debug       : NULL
..@ inherit.blank: logi TRUE
$ axis.text.x.top  : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA
..@ colour      : NULL
..@ size        : NULL
..@ hjust       : NULL
..@ vjust       : num 0
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : <ggplot2::margin> num [1:4] 0 0 2.4 0
..@ debug       : NULL
..@ inherit.blank: logi TRUE
$ axis.text.x.bottom : NULL
$ axis.text.y      : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA

```

```

..@ colour      : NULL
..@ size        : NULL
..@ hjust       : num 1
..@ vjust       : NULL
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : <ggplot2::margin> num [1:4] 0 2.4 0 0
..@ debug       : NULL
..@ inherit.blank: logi TRUE
$ axis.text.y.left      : NULL
$ axis.text.y.right     : <ggplot2::element_text>
..@ family            : NULL
..@ face              : NULL
..@ italic            : chr NA
..@ fontweight        : num NA
..@ fontwidth         : num NA
..@ colour            : NULL
..@ size              : NULL
..@ hjust             : num 0
..@ vjust             : NULL
..@ angle             : NULL
..@ lineheight        : NULL
..@ margin            : <ggplot2::margin> num [1:4] 0 0 0 2.4
..@ debug             : NULL
..@ inherit.blank     : logi TRUE
$ axis.text.theta       : NULL
$ axis.text.r           : <ggplot2::element_text>
..@ family            : NULL
..@ face              : NULL
..@ italic            : chr NA
..@ fontweight        : num NA
..@ fontwidth         : num NA
..@ colour            : NULL
..@ size              : NULL
..@ hjust             : num 0.5
..@ vjust             : NULL
..@ angle             : NULL
..@ lineheight        : NULL
..@ margin            : <ggplot2::margin> num [1:4] 0 2.4 0 2.4
..@ debug             : NULL
..@ inherit.blank     : logi TRUE
$ axis.ticks            : <ggplot2::element_line>
..@ colour            : chr "#333333FF"

```

```

..@ linewidth      : NULL
..@ linetype       : NULL
..@ lineend        : NULL
..@ linejoin       : NULL
..@ arrow          : logi FALSE
..@ arrow.fill     : chr "#333333FF"
..@ inherit.blank: logi TRUE
$ axis.ticks.x      : NULL
$ axis.ticks.x.top  : NULL
$ axis.ticks.x.bottom : NULL
$ axis.ticks.y      : NULL
$ axis.ticks.y.left : NULL
$ axis.ticks.y.right : NULL
$ axis.ticks.theta  : NULL
$ axis.ticks.r       : NULL
$ axis.minor.ticks.x.top : NULL
$ axis.minor.ticks.x.bottom : NULL
$ axis.minor.ticks.y.left : NULL
$ axis.minor.ticks.y.right : NULL
$ axis.minor.ticks.theta : NULL
$ axis.minor.ticks.r : NULL
$ axis.ticks.length : 'rel' num 0.5
$ axis.ticks.length.x : NULL
$ axis.ticks.length.x.top : NULL
$ axis.ticks.length.x.bottom : NULL
$ axis.ticks.length.y : NULL
$ axis.ticks.length.y.left : NULL
$ axis.ticks.length.y.right : NULL
$ axis.ticks.length.theta : NULL
$ axis.ticks.length.r : NULL
$ axis.minor.ticks.length : 'rel' num 0.75
$ axis.minor.ticks.length.x : NULL
$ axis.minor.ticks.length.x.top : NULL
$ axis.minor.ticks.length.x.bottom : NULL
$ axis.minor.ticks.length.y : NULL
$ axis.minor.ticks.length.y.left : NULL
$ axis.minor.ticks.length.y.right : NULL
$ axis.minor.ticks.length.theta : NULL
$ axis.minor.ticks.length.r : NULL
$ axis.line         : <ggplot2::element_blank>
$ axis.line.x       : NULL
$ axis.line.x.top   : NULL
$ axis.line.x.bottom : NULL

```

```

$ axis.line.y           : NULL
$ axis.line.y.left     : NULL
$ axis.line.y.right    : NULL
$ axis.line.theta      : NULL
$ axis.line.r          : NULL
$ legend.background    : <ggplot2::element_rect>
  ..@ fill             : chr "white"
  ..@ colour           : logi NA
  ..@ linewidth        : NULL
  ..@ linetype         : NULL
  ..@ linejoin         : NULL
  ..@ inherit.blank    : logi FALSE
$ legend.margin        : NULL
$ legend.spacing       : 'rel' num 2
$ legend.spacing.x     : NULL
$ legend.spacing.y     : NULL
$ legend.key           : NULL
$ legend.key.size      : 'simpleUnit' num 1.2lines
  ..- attr(*, "unit")= int 3
$ legend.key.height    : NULL
$ legend.key.width     : NULL
$ legend.key.spacing   : NULL
$ legend.key.spacing.x : NULL
$ legend.key.spacing.y : NULL
$ legend.key.justification : NULL
$ legend.frame         : NULL
$ legend.ticks         : NULL
$ legend.ticks.length  : 'rel' num 0.2
$ legend.axis.line     : NULL
$ legend.text          : <ggplot2::element_text>
  ..@ family           : NULL
  ..@ face             : NULL
  ..@ italic           : chr NA
  ..@ fontweight       : num NA
  ..@ fontwidth       : num NA
  ..@ colour           : NULL
  ..@ size             : 'rel' num 0.8
  ..@ hjust            : NULL
  ..@ vjust            : NULL
  ..@ angle            : NULL
  ..@ lineheight       : NULL
  ..@ margin           : NULL
  ..@ debug            : NULL

```

```

..@ inherit.blank: logi TRUE
$ legend.text.position      : NULL
$ legend.title              : <ggplot2::element_text>
..@ family                  : NULL
..@ face                    : NULL
..@ italic                  : chr NA
..@ fontweight              : num NA
..@ fontwidth               : num NA
..@ colour                  : NULL
..@ size                    : NULL
..@ hjust                   : num 0
..@ vjust                   : NULL
..@ angle                   : NULL
..@ lineheight              : NULL
..@ margin                  : NULL
..@ debug                   : NULL
..@ inherit.blank: logi TRUE
$ legend.title.position     : NULL
$ legend.position           : chr "top"
$ legend.position.inside    : NULL
$ legend.direction          : NULL
$ legend.byrow              : NULL
$ legend.justification      : chr "center"
$ legend.justification.top   : NULL
$ legend.justification.bottom : NULL
$ legend.justification.left  : NULL
$ legend.justification.right : NULL
$ legend.justification.inside : NULL
[list output truncated]
@ complete: logi TRUE
@ validate: logi TRUE

```

```
ggsave("results/SWYC/development_quotient.png")
```

Saving 5.5 x 3.5 in image

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

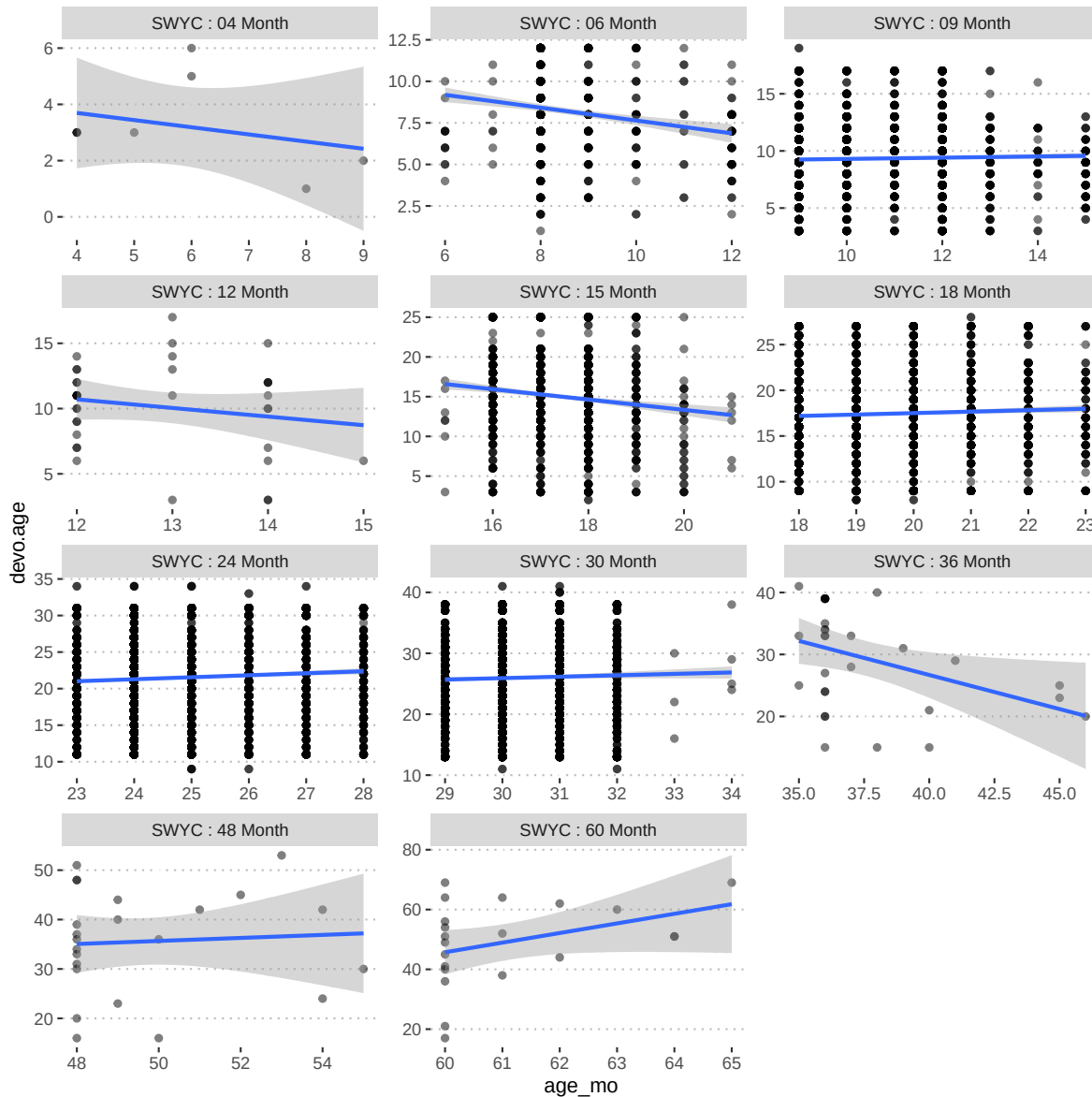
The devo.quotient is below 1 (mean = 0.9), but I suppose that is to be expected in a cohort composed entirely of young children with brain MRIs.

Visualize Developmental Quotient Trends

Let's also verify that the computed developmental age tracks positively with chronological age across the different form windows.

```
ggplot(qforms.swyc.indiv, aes(x = age_mo, y = devo.age)) +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "lm") +  
  theme_pubclean() +  
  facet_wrap(~ SWYC.form_friendly_name, ncol = 3, scales = "free")
```

`geom\_smooth()` using formula = 'y ~ x'



```
ggsave("results/SWYC/swyc_vs_devo_age.png")
```

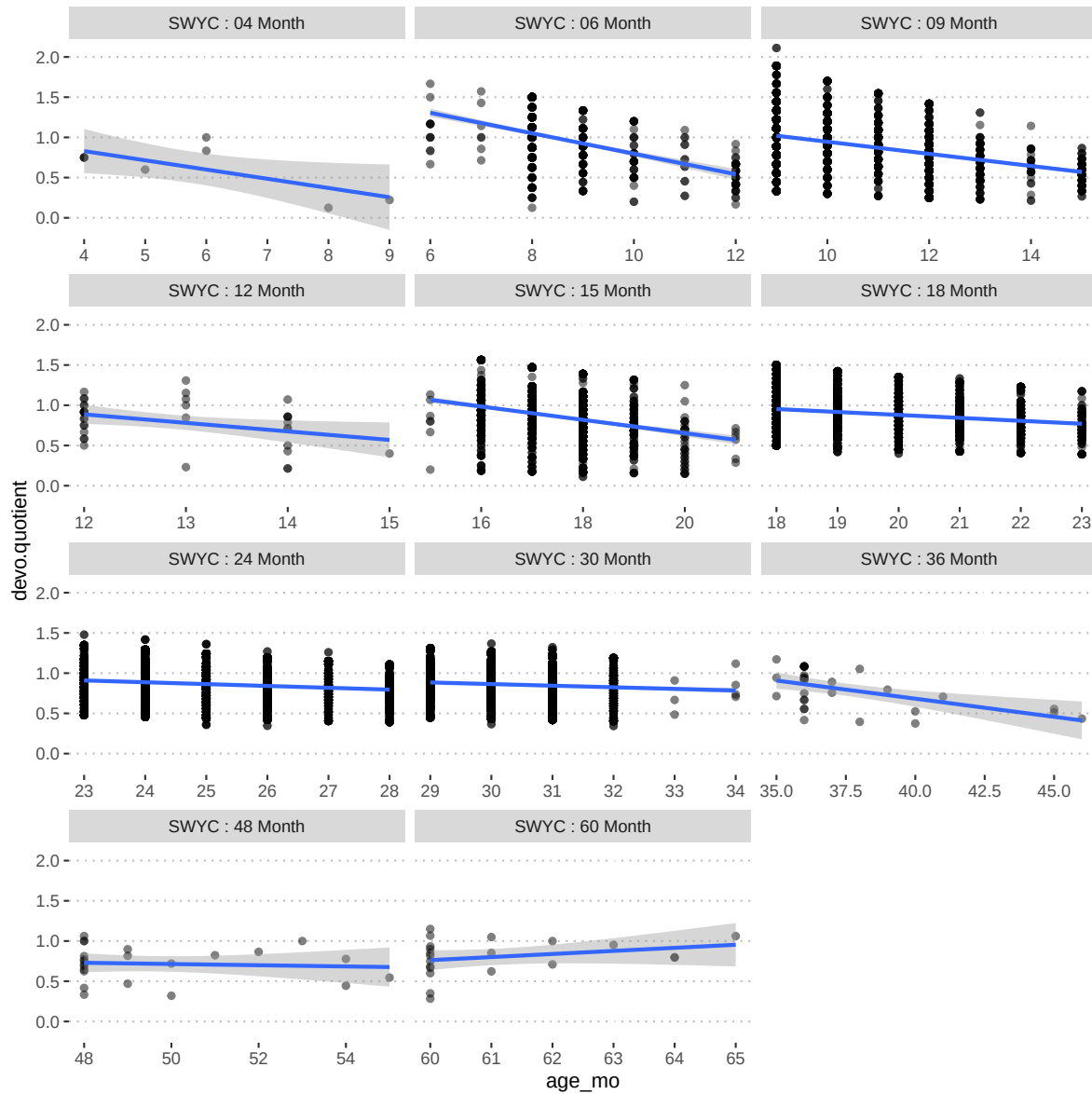
Saving 9 x 9 in image

`geom\_smooth()` using formula = 'y ~ x'

```
ggplot(qforms.swyc.indiv, aes(x = age_mo, y = devo.quotient)) +  
  geom_point(alpha = 0.5) +
```

```
geom_smooth(method = "lm") +  
theme_pubclean() +  
facet_wrap(~ SWYC.form_friendly_name, ncol = 3, scales = "free_x")
```

`geom\_smooth()` using formula = 'y ~ x'



```
ggsave("results/SWYC/swyc_vs_dq.png")
```

Saving 9 x 9 in image

```
`geom_smooth()` using formula = 'y ~ x'
```

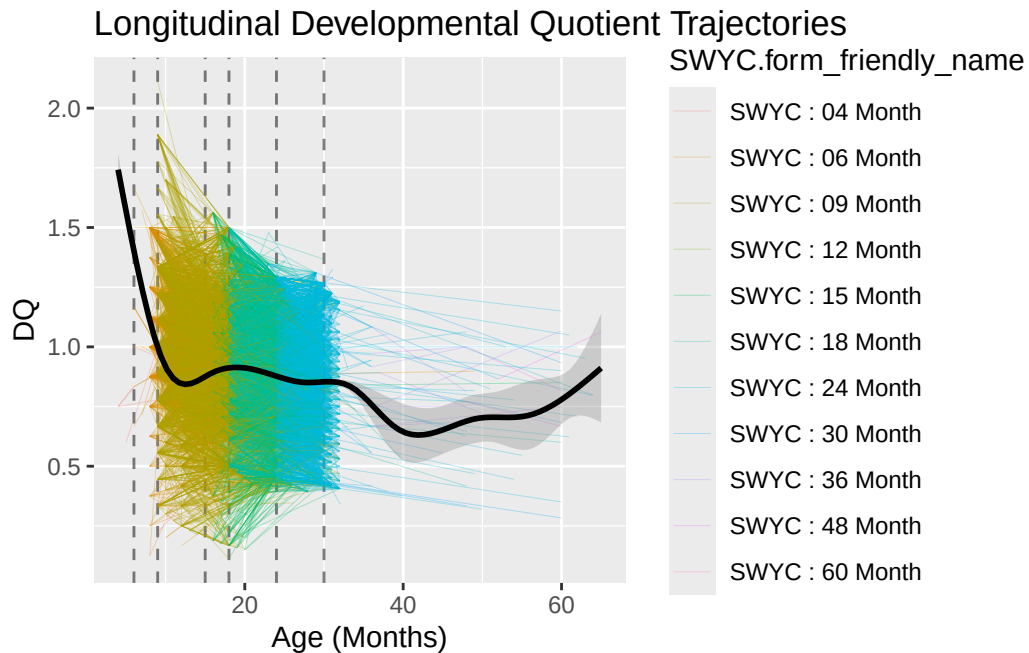
Developmental ages are relatively flat with actual age and developmental quotient tends to decrease with age within the same assessments. We'd expect developmental age to increase with actual age, and we'd expect developmental quotient to remain relatively flat. Perhaps this reflects ascertainment bias. Individuals with assessments closer to the age threshold may reflect regular wellness assessments, whereas later assessments may reflect physicians suspecting developmental delay.

Visualize Developmental Quotient Longitudinal Data

Finally, let's visualize the longitudinal data to see how individuals traverse the milestones over multiple well-child visits.

```
ggplot(qforms.swyc.indiv, aes(x = age_mo, y = devo.quotient, group = pat_id, color = SWYC.for  
  geom_vline(xintercept = c(6, 9, 15, 18, 24, 30), linetype = "dashed", alpha = 0.5) +  
  geom_line(size = 0.05, alpha = 0.3) +  
  stat_smooth(aes(group = 1), color = "black", size = 1) +  
  labs(title = "Longitudinal Developmental Quotient Trajectories", x = "Age (Months)", y = "I
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Save

```
write_csv(qforms.swyc.indiv, "results/SWYC/swyc_normative_scores.csv")
```

Upload To Arcus

Push the fully cleaned and processed normative scores back into the BigQuery lab space.

```
# Clean up table naming conventions for BigQuery schema compliance
df_swyc <- qforms.swyc.indiv %>%
  select(pat_id, encounter_id, sex_abbr, effective_age,
         age_mo, gestational_age_num, weeks_premature,
         prematurity_adj_age = clin.prematurity_adj_age,
         form_friendly_name = SWYC.form_friendly_name,
         n_answers = SWYC.n_answers,
         sum_answers = SWYC.sum_answers,
         all_0s = SWYC.all_0s,
         all_2s = SWYC.all_2s,
         meets_milestones = clin.meets_milestones,
```

```

        norm_devo_age = devo.age,
        norm_devo_quotient = devo.quotient,
        norm_devo_prob = devo.prob,
        appt_prc_name)

colnames(df_swyc) <- str_replace_all(colnames(df_swyc), "\\.", "_")

```

```

# Define target table
bq_swyc <- bq_table(project_id, "lab", paste0("swyc_normative_scores_", Sys.Date()))

# Refresh the table if it already exists
if (bq_table_exists(bq_swyc)) {
  bq_table_delete(bq_swyc)
}

# Apply metadata to the table for data governance and reproducibility
bq_table_create(
  bq_swyc,
  fields = df_swyc,
  friendly_name = "SWYC Normative Scores",
  description = "These data contain summarized and normative scores for the developmental mi.
)

```

```
<bq_table> scit605-healthyMRI-2d73d69f.lab.swyc_normative_scores_2026-02-24
```

```
bq_table_exists(bq_swyc)
```

```
[1] TRUE
```

```

# Execute upload
bq_table_upload(bq_swyc, df_swyc)

bq_table_fields(bq_swyc)

```

```

<bq_fields>
  pat_id <STRING>
  encounter_id <STRING>
  sex_abbr <STRING>
  effective_age <INTEGER>
  age_mo <FLOAT>

```

gestational_age_num <STRING>
weeks_premature <FLOAT>
prematurity_adj_age <FLOAT>
form_friendly_name <STRING>
n_answers <INTEGER>
sum_answers <FLOAT>
all_0s <BOOLEAN>
all_2s <BOOLEAN>
meets_milestones <BOOLEAN>
norm_devo_age <FLOAT>
norm_devo_quotient <FLOAT>
norm_devo_prob <FLOAT>
appt_prc_name <STRING>